Encoding (a Short History)

Chapter 2, "Data, Types, and Variables," began with an example of how string data is represented in memory by a sequence of bytes where each byte value was mapped to a particular character. This equivalence between value and character is called *encoding*. Encoding enables machines to transfer (and store) text characters, and Morse code is one of the earliest examples of such a system: four symbols (long and short dash, long and short space) were initially used to encode the English alphabet, digits *0* to *9*, some limited punctuation, and a handful of control sequences (called "prosigns"). Other examples of early encoding schemes include Braille, signal flags on ships, and so on. Encoding requires that the sender and receiver agree on what the signals mean, which of course requires standards that define them. The story of the evolution of character encoding standards on computer systems is a long one and only of interest to certain types of people (I am absolutely one of those types of people, but recognize I'm in the minority), so we'll skip towards the end…

ASCII

The basis for modern electronic character encoding began with the American National Standards Institute (ANSI) and their ASCII63 (American Standard Code for Information Interchange, 1963) standard, which eventually saw wide adoption (prior to ASCII63, encoding was highly vendor-specific, as indeed it continued to be for many years after 1963). The ASCII system uses seven bits to store the numerical value of the character, which means there are 128 available *codepoints*. Codepoints are numbers that are assigned to characters within a *code page* (sometimes called a *code space*), where the code page is the full set of characters. (Fun fact: we get the term "code page" from the days when system technical manuals would have the character mappings printed out for lookup – a literal page in a book.) These days other terms exist, such as *character set*, *charset*, or *charmap*; they all mean the same thing: a set of characters mapped to numeric values (either specific sequences of bits or ordinal numbers). Why seven bits? Well, this standard dates from a time when an extra bit would have represented significantly increased costs in data transfer and storage. An IBM Model 350 hard disk in 1956 stored 3.75 megabytes, weighed over a ton, and could only be leased – not bought – for around $3000 a *month*!